

Waiting for 3G?

KeyOn Communications Holdings Inc. and Alvarion have deployed a 4G wireless network in Nevada in the last week of July 2010

Goodbye shopping woes

Carbon nanotube-laced ink will soon be used to print RFID tags on goods in shopping malls, enabling you to simply walk out of the mall with your purchases. Wonder when we'll get to see the technology at work!

Tomorrow's tech

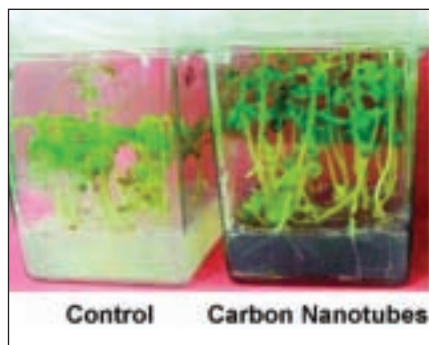
thick outer shell and bonding with the plant structure that enable the plants to take in more water and also retain it, creating a plant that could possibly survive longer spells of drought after long rain fall.

Future applications

Although current applications may be limited by the prohibitive cost of \$95 per gram (that's a little under Rs. 4,500) at present. Probably with time, it would become more affordable.

Hydrogen storage

Carbon nanotubes have, for long, been championed as a strong contender for storing hydrogen in fuel cells.



Accelerating food growth with the aid of carbon nanotubes

Improved transistors

Y-shaped nanotubes behaving as electronic switches similar to traditional MOS (metal oxide semiconductor) transistors currently used in almost all of microprocessors, memory, and other integrated circuits have been invented by researchers at UCSD and Clemson University. They say that these specially synthesised carbon nanotube structures exhibit electronic properties that are improved over conventional transistors used in computers.

Water filter

At our very own Bhabha Atomic Research Centre (BARC) in Mumbai new water purification technologies are constantly being investigated that are simple, inexpensive to install, operate, and maintain. Their recent efforts in the field of nanotechnology has led them to use hollow carbon nanotubes to do the job of water purification. The unique chemical properties of carbon nanotubes

mean that only very small molecules, such as water molecules can pass along their interiors, whereas viruses, bacteria, toxic metal ions, and large noxious organic molecules cannot.

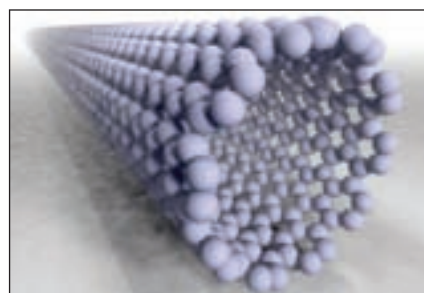
Smooth and water repellent surfaces of carbon nanotubes means that a filter-based on this technology would be very efficient, allowing a high flow rate of water through the filter without fouling. Importantly, the power needed to drive water through such a system will be low compared to conventional membrane technology. Although this technology is not without limitations, in order to be a practical solution, the nanoscale structures need to be engineered to form well-defined arrangements to allow the efficient decontamination of water.

Breast cancer research

Nanotubes bond to an antibody that is produced by chickens and have been useful in lab tests to destroy breast cancer tumors. The antibody carrying nanotubes are attracted to proteins produced by a type of breast cancer cell. Then the nanotubes absorb light from an infrared laser, literally burning the nanotube and the breast cancer cell they are attached to.

Single-walled carbon nanotubes have been highly touted for their potential as novel delivery agents for cancer detection and therapeutic agents. HER2 is a family of genes that help regulate the growth and proliferation of human cells. Normal cells have two copies of HER2, but about 20 to 25 per cent of breast cancer cases consist of cells with multiple copies of the gene, resulting in the overproduction of a HER2-encoded protein that is associated with particularly fast growing tumours.

The team led by Huixin He, PhD, of Rutgers University, and Yan Xiao, PhD,

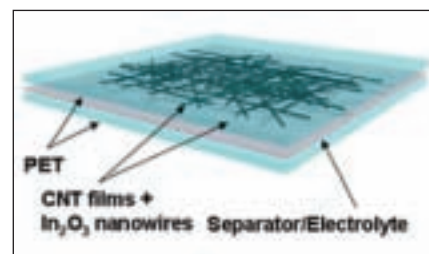


Hollow Carbon Nanotubes like these can be used to filter water

of the National Institute of Standards and Technology (NIST), has demonstrated how they created the new dual-purpose nano structure by attaching an anti-HER2 antibody to short carbon nanotubes. The investigators then exploited two unique optical properties of carbon nanotubes to detect and then destroy HER2 breast cancer cells. Near-infrared laser light at a wavelength of 785 nanometres reflects intensely off the nanotubes, and this strong signal is easily detected by a technique called Raman spectroscopy (invented by Dr. C V Raman). Increase the laser light's wavelength to 808 nanometres and it will be absorbed by the nanotubes, burning them and anything to which they're attached, in this case, the HER2 tumour cells.

Battery

The latest iPhone4 has more than half of its volume occupied by the mammoth battery which is crucial to running the modern convergent devices with power hungry features such as GPS, Wi-fi, powerful CPUs and large screens. Battery is currently the biggest hurdle in making better and more featureful mobile devices (iPhone didn't support multitasking till now to primarily save on battery). All this is about to change



They might power your next cellphone

with MIT's Laboratory for Electromagnetic and Electronic Systems (LEES) work on technologically possible and economically viable carbon nanotube based ultra-capacitor alternative.

Capacitors store energy as an electrical field, making them more efficient than standard batteries, which get their energy from chemical reactions. Ultra-capacitors are capacitor-based storage cells that provide instant, massive bursts of instant energy. They have already been used in fuel-cell vehicles to provide an extra burst for accelerating into traffic and climbing hills. 